⇒ Unbounded Search

- we don't know the end!
- ∞ array

→ linear search (sorted arr)

```
↳ while (true)
  {
    if (arr[i] > x)
    {   break;
    }
    if (arr[i] == x)
    {   ans = i;
    }
    i++;
  }
```

$\boxed{n = 6}$

index	0	1	2	3	4	5	6	7	8	9	10 . . . .
⇒	0	1	2	3	4	5	6	7	8	9	10 . . . . .
	~~i~~	i ~~j~~	~~j~~ ~~i~~		~~j~~ i				j		

```
while ( a[j] < x) → 8 ≤ 6 X
  {
      i = j
      j = j * 2;
  }
```

BS (arr, i , j , x)